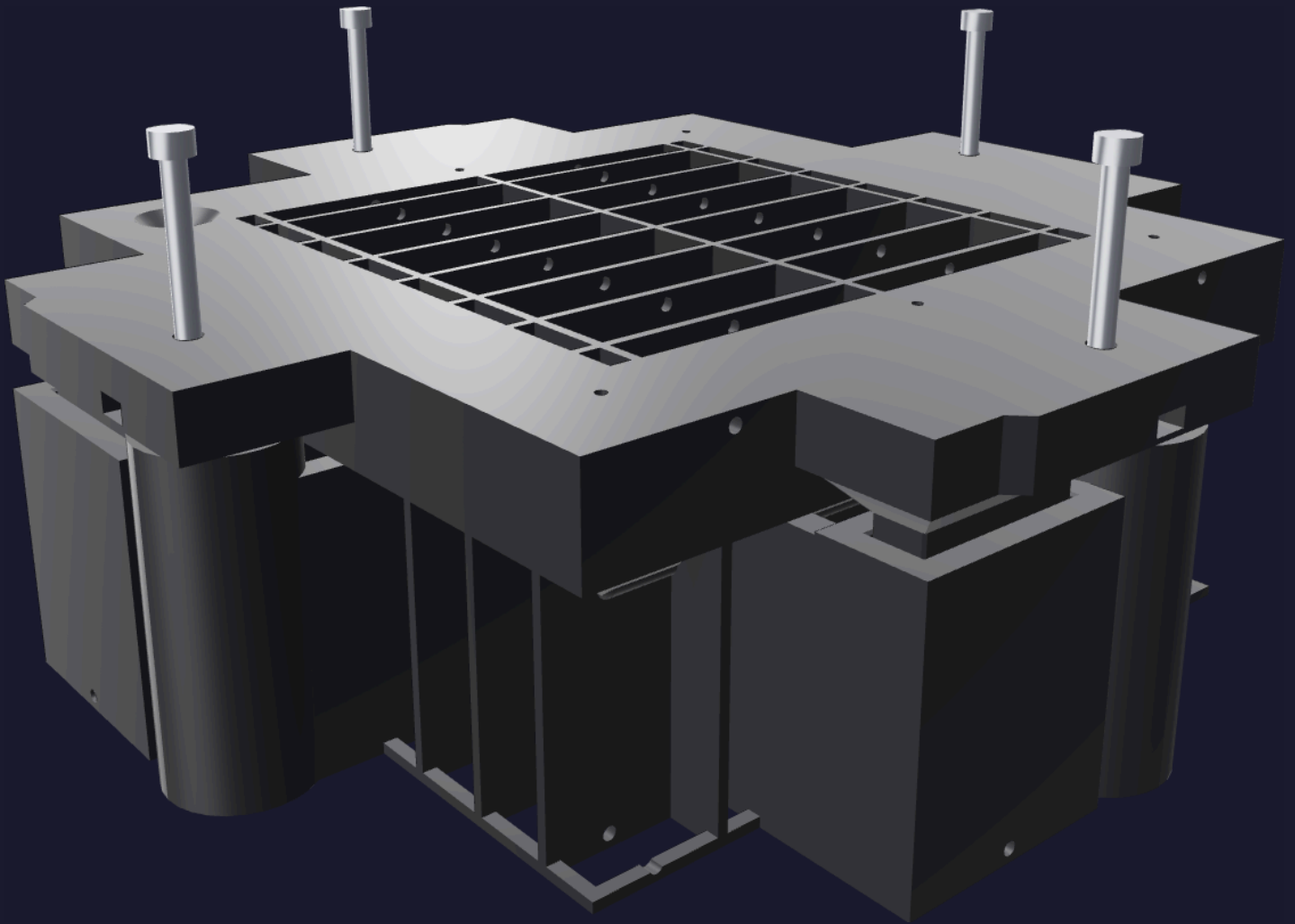


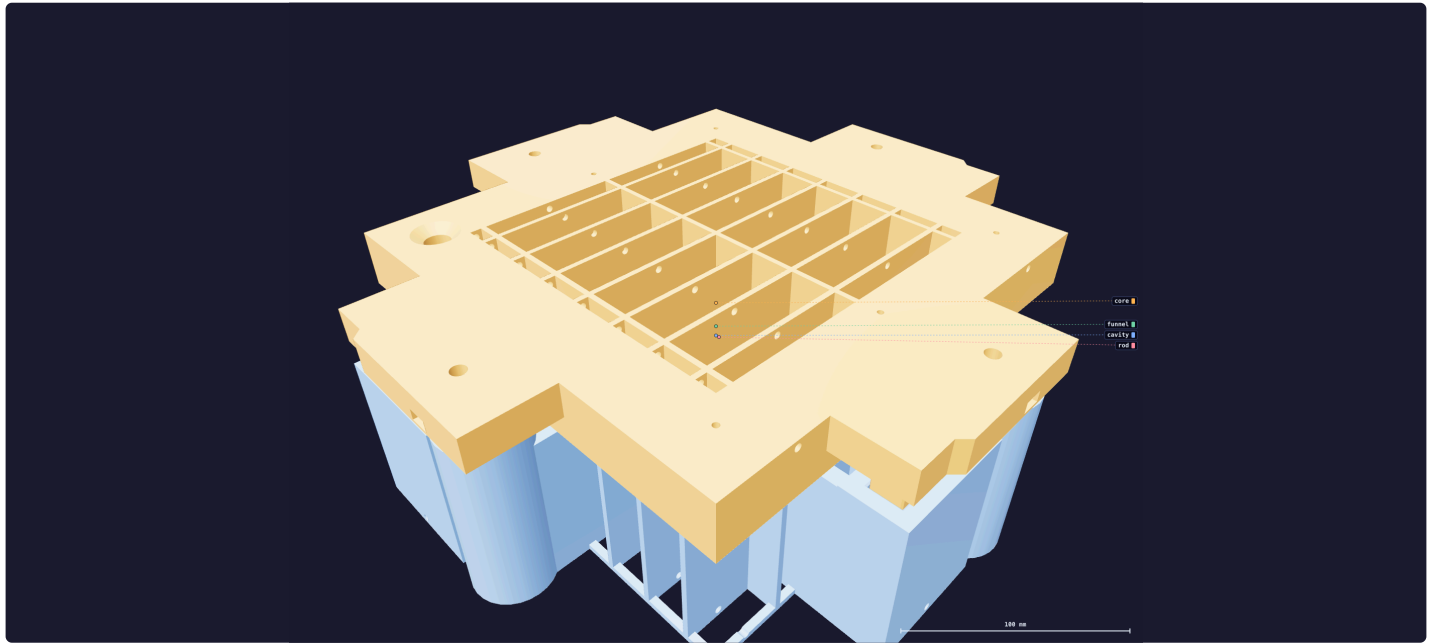
# Funnel mold

Two printed PETG halves that cast the Zone C silicone funnel around a ground steel dowel. Three prints, one finishing process, four steel jacks.



# What you are building

A two-part mold. The **cavity** forms the outside of the funnel and is poured into. The **core** forms the inside and drops onto it. A ground steel dowel forms the spout bore. Four screw jacks lift the core off the cured silicone.



Closed on the pour, cut at the parting plane. **Cavity** below, **core** above, the **silicone** between them, the **rod** standing in the spout. The mold is 270 × 270 mm square.

## Two documents

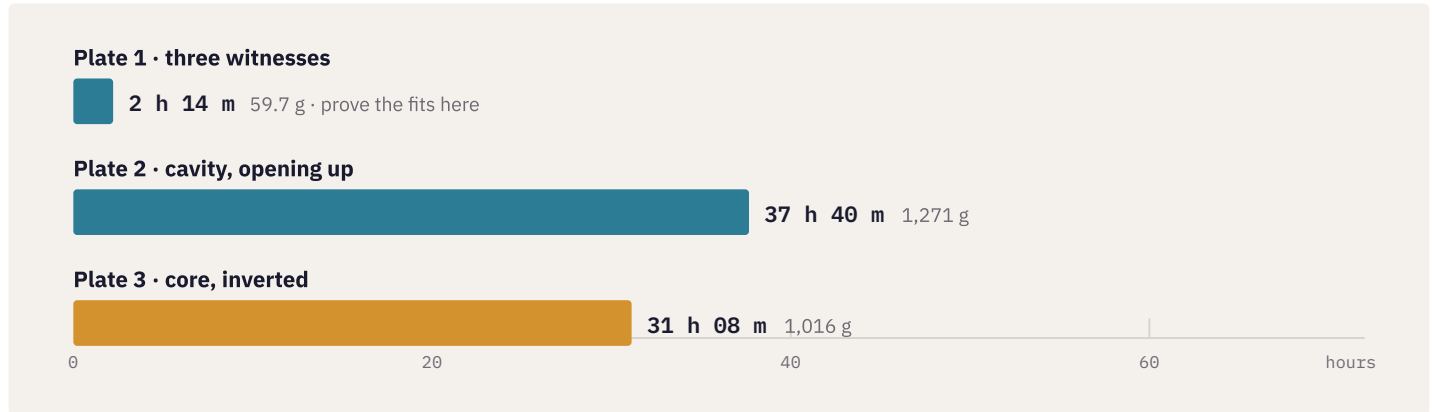
This one builds the mold: three print jobs, a finishing process measured on a coupon, four steel stations fitted. It ends with a tool you keep. Its companion, **Casting a funnel**, is the one you take to the bench with gloves on — it runs on a clock, because mixed silicone has a working time.

### BEFORE YOU START

Read this document through, once, end to end. Two of the three prints are longer than a working day and the fits that decide them are checked on page 6.

# The whole path

Six phases. Nearly three days of them are the printer running unattended, and the two long plates are only started after a two-hour plate has proved the fits they depend on.



Sliced times from the delivered project, to scale. Each plate is an independent job. Together they are **71 hours** of machine time and **2,347 g** of PETG.

- 1 Print the witnesses and check the fits.** Two hours buys you the right to start a 37-hour plate.
- 2 Print the cavity, then the core.** Clear every air passage as each comes off.
- 3 Dry-assemble.** Full stroke, real rod, before any coating.
- 4 Prove the finishing process on the coupon.** Then apply it to the two forming faces.
- 5 Fit the steel.** Four nuts, four washers, four screws.
- 6 Dry-run the empty mold in the chamber.**

# What the mold is made of

Everything here is in the shop except the jack screws, which are on order. Check them off before you dry filament — the finishing coat is the only line with an open decision on it, and page 15 is where you close it.

| ITEM           | SPEC                                                                                                                                                                     | QTY          | WHERE    |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------|
| PETG filament  | Bambu <b>PETG Translucent</b> , 1.75 mm. Dry it: <b>65 °C</b> for <b>8 h</b> minimum.                                                                                    | <b>~3 kg</b> | owned    |
| Jack screws    | M5 × 0.8 × <b>50 mm</b> fully threaded socket head, 12.9 alloy. <b>The 4 mm hex key in the pack is the tool you demold with.</b>                                         | <b>4</b>     | on order |
| Square nuts    | M5 × 0.8, <b>8 mm</b> square × <b>4 mm</b> thick, stainless                                                                                                              | <b>4</b>     | owned    |
| Fender washers | M5, <b>25 mm</b> OD steel. The pocket takes <b>0.8–2.0 mm</b> thickness — <b>measure your stock.</b>                                                                     | <b>4</b>     | owned    |
| Spout rod      | Ground steel dowel, <b>6.35 × 50.8 mm</b> ( $\frac{1}{4}'' \times 2''$ ), ground to 0.0001"–0.0005". <b>This is the funnel's whole bore.</b>                             | <b>1</b>     | owned    |
| Finishing coat | Candidates in the shop: <b>Krylon Crystal Clear acrylic</b> (gloss) and <b>Craft Resin</b> crystal-clear epoxy. XTC-3D is the purpose-made alternative and is not owned. | <b>1</b>     | choose   |
| Release agent  | Mann <b>Ease Release 200</b> , addition-cure compatible. A release film, not a silicone fluid.                                                                           | <b>1</b>     | owned    |

## 🚫 BOTH LARGE PLATES OUTFRAN A SPOOL

The cavity is **1,271 g** and the core **1,016 g**, on 1 kg spools. Arrange automatic spool backup, or plan a supervised runout change, **before** you start either plate. The saved project does not configure the printer's backup spools for you.

# Set the printer

Open **funnel-mold-guided-vacuum-petg-08-016.3mf** in Bambu Studio. It already carries three sliced plates, the printer's Z trim and the speed modifiers. You are confirming the machine matches it, not re-deciding it.

| SETTING     | VALUE                                      | SETTING      | VALUE              |
|-------------|--------------------------------------------|--------------|--------------------|
| Machine     | <b>H2C</b>                                 | Layer height | <b>0.16 mm</b>     |
| Nozzle      | <b>0.8 mm Standard, LEFT</b>               | First layer  | <b>0.30 mm</b>     |
| Filament    | PETG Translucent, <b>Left Extruder (1)</b> | Walls        | <b>4 × 0.80 mm</b> |
| Nozzle temp | <b>255 °C</b>                              | Infill       | <b>100 %</b>       |
| Plate       | Textured PEI, <b>70 °C</b>                 | Supports     | <b>off</b>         |

## The two speed modifiers

Each large plate carries a modifier volume named *Forming faces, registration and hardware fits*. Inside it the outer wall drops from **100** to **30 mm/s**. It covers the forming skins, the registration lands, the rod boss, the guide blades and bearings, the nut seats and the washer pockets. The exposed backing ribs stay fast. Do not flatten it to one speed to save an hour.

### CONFIRM, THEN GO

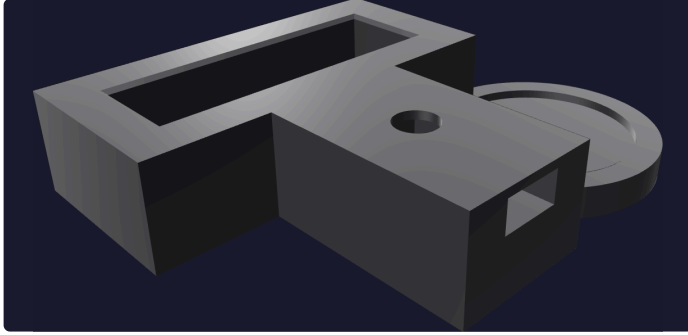
If Studio asks to group filaments, keep **Custom** with PETG on Left Extruder (1). Re-confirm the 0.8 mm Standard nozzle after any hardware sync. Re-slice only if you change the printer, nozzle, trim, filament, geometry or the finishing allowance.

### ⚠ THE PLATE IS HOT AND THE JOB IS LONG

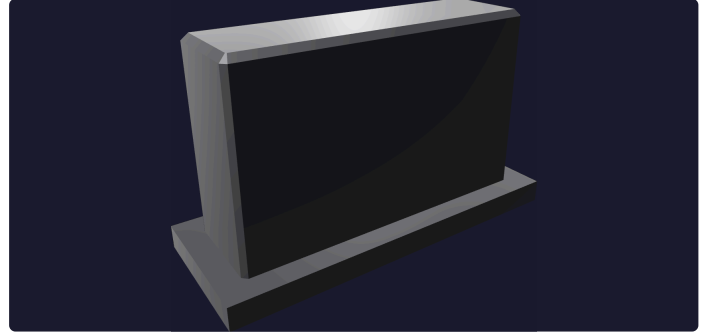
Textured PEI at 70 °C for 37 hours. Leave the enclosure closed while it runs, and let each plate cool **on the bed** before you lift anything off it.

# Three witnesses first

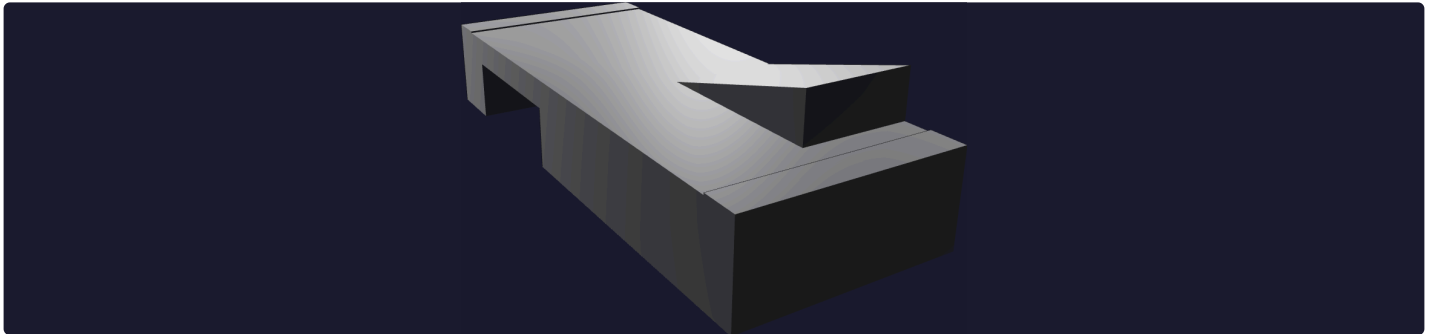
Plate 1 is **2 h 14 m** and **59.7 g**. It carries three coupons that reproduce, at full size and in the real print orientation, every fit the two long plates depend on. Print it, check it, correct it — then start the 37-hour plate.



**Hardware witness.** The nut slot, a guide bearing and a washer pocket.



**Guide blade.** A **24 mm** free length of the real blade.



**Finish witness.** A flat measuring strip between two rails, a ramp at the funnel's own floor grade, and a **17.6 mm** bridge window underneath at the mold's rib spacing.

- 1 **Print plate 1 alone.** Same nozzle, same filament, same 0.16 mm layers as the mold. All three coupons carry the 30 mm/s override.
- 2 **Let it cool on the bed, then remove the brim and every string.** A string in a bore reads as a tight fit that isn't one.
- 3 **Check the three fits on page 7 and the coupon on page 8.**

## DONE WHEN

All three coupons are off the plate, deburred, and every fit on the next two pages passes without force.

# Check the three fits

Do these with the actual steel you will build with, not with a nominally identical box of it. Every one of them must go together **by hand, without force and without rocking**. A fit you have to persuade on a 60 mm coupon is a fit you cannot persuade on a 270 mm plate.

- A The square nut into its slot.** Slide one nut in from the block's outside edge. The slot is **8.4 × 4.4 mm** on an **8 × 4 mm** nut. Then thread an M5 screw through the **5.8 mm** bore into it and back it out again. The slot is reproduced in the *core's inverted print orientation*, so what you are really checking is the thin retaining lip that bridges over it.
- B The blade through the bearing.** Slide the guide-blade coupon through the sleeve in the hardware witness — **12 × 40 mm** blade in a **12.6 × 40.6 mm** bore. It must slide freely and must not rock. This is the fit that decides whether the core comes off straight.
- C The washer into its pocket.** Drop one fender washer flat into the **25.4 mm** pocket. It must sit on a flat, fully supported floor with no rock. **Measure the washer's thickness now** and write it down; the design accommodates **0.8–2.0 mm**.
- D The bridge roof on the finish witness.** Look underneath at the **17.6 mm** window. The roof over it must be attached at *both* ends. That same span carries the forming skin across every rib bay in the mold.

## ⊖ CORRECT IT HERE OR NOT AT ALL

A tight fit is fixed on this coupon — by clearing strings, adjusting flow, or re-slicing — and then re-printed as a 2-hour plate. Discovering it after 69 hours of printing costs you both large parts.

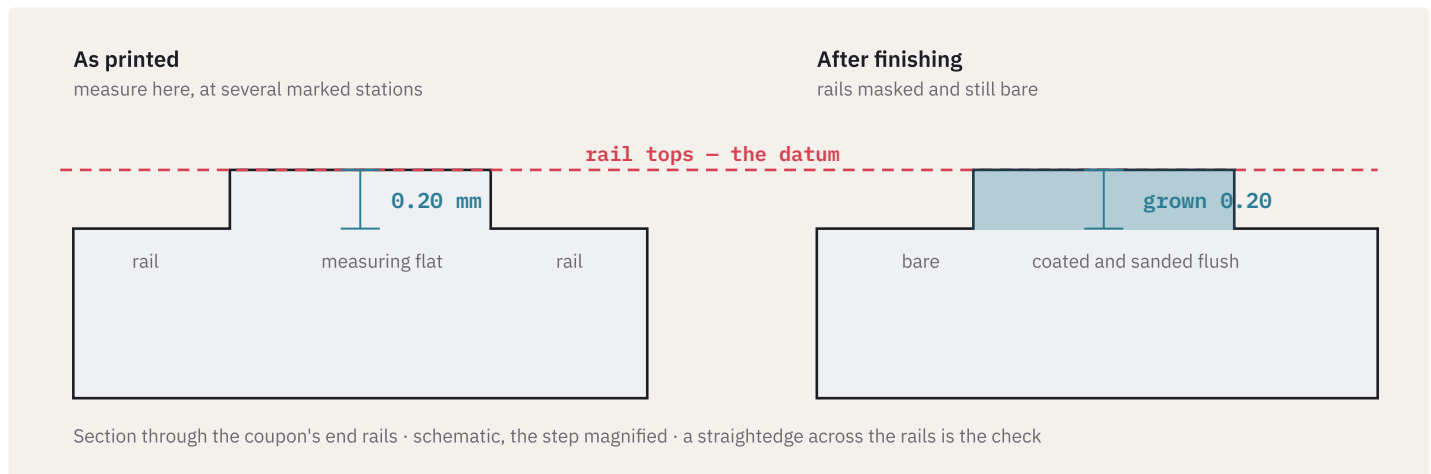
What this coupon cannot tell you: whether the four widely spaced guides on the real core line up with the four bearings on the real cavity. That check is the dry assembly on page 13.

## DONE WHEN

Nut, screw, blade and washer each go home by hand, and the bridge roof is sound at both ends. Washer thickness recorded.

# Measure the bare coupon

The finish witness is a gauge for one number: **0.20 mm** of net growth. Its two end rails stand exactly that far above its measuring flat, so when the coated flat comes up level with the rails, the process is on target. Measure it **before** you coat anything.



The rails are the zero. Everything you do to the flat is read as a **difference** from them, which is what makes a caliper good enough for a  $\pm 0.05$  mm target.

- 1 Mark stations along the flat** and record the uncoated thickness at each with calipers. Several — one reading is not a measurement.
- 2 Measure the actual printed rail step too.** It is modelled at 0.20 mm, but layer quantisation and first-layer error move it. The number you write down is the gauge, not the number in the CAD.
- 3 Keep the rails and the underside bare** from here on. They are the datum and the bridge inspection window.
- 4 Use the ramp as the honest half of the test.** It is cut at the funnel's own floor grade, so it asks the question a flat cannot: does your process take the layer terraces off a shallow slope *without* rounding the transitions at the top and bottom of it, and without the coating pooling at the foot of the grade?

## TWO NUMBERS, NOT ONE

A spray builds differently on a floor than on a wall. Measure a vertical face separately from the flat and record both — the mold has far more vertical forming surface than horizontal.

## DONE WHEN

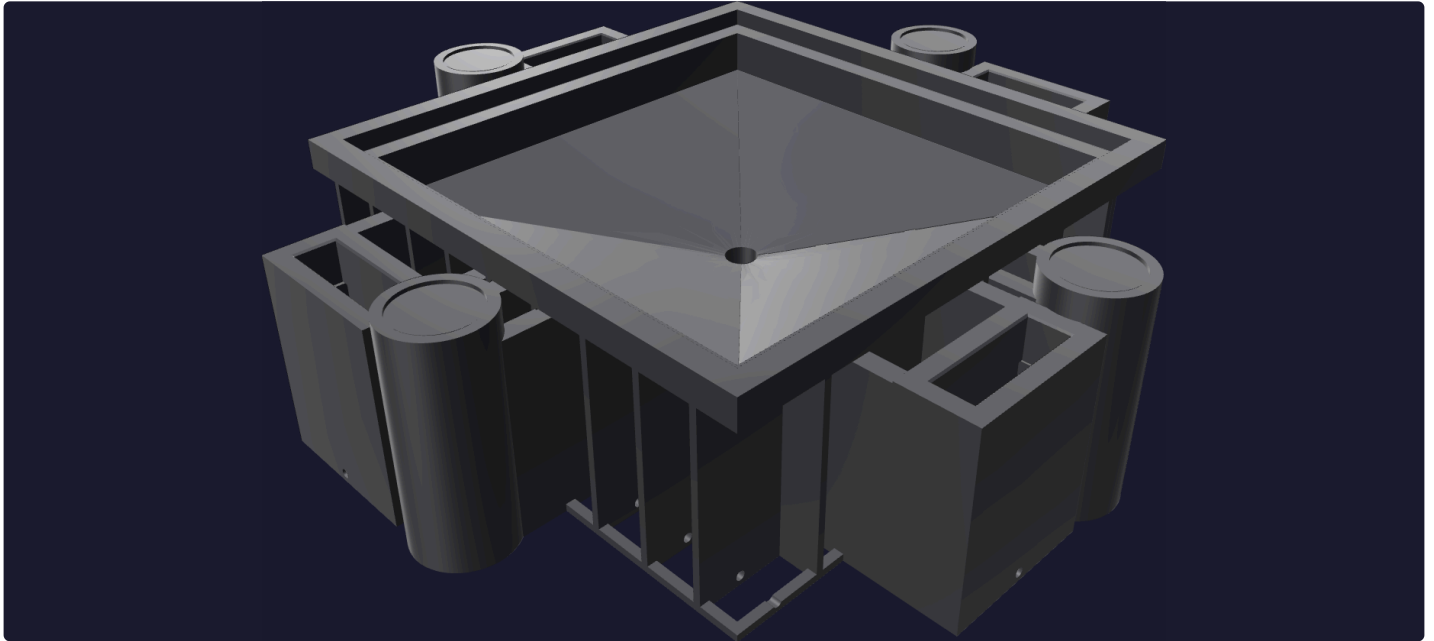
You have written down the bare flat thickness at each marked station and the real rail step. Nothing is coated yet.



# The cavity

The half you pour into. It prints **opening up**, exactly as it sits on the bench, so the forming skin is built on ribs and the whole backing structure opens downward to the bed.

37 h 40 m · 1,271 g · 466 layers .



Off the bed, right way up. The **forming skin** falls from the registration band to the blind spout pocket at the centre. Four round **bearing columns** carry the washer pockets; four rectangular **guide towers** take the core's blades. Everything between the ribs underneath is open air.

## What the structure is for

The skin that forms the funnel is only **3.2 mm** thick, carried on **2.4 mm** ribs at **20 mm** centres. The bays between those ribs are not infill the slicer skipped — they are modelled open volume, and transverse **3 mm** channels connect them to the outside so chamber pressure reaches them. A printed mold is porous; the ribs and channels equalise it, and the coating is what actually seals it.

### ⚠ PLAN THE SPOOL BEFORE YOU PRESS GO

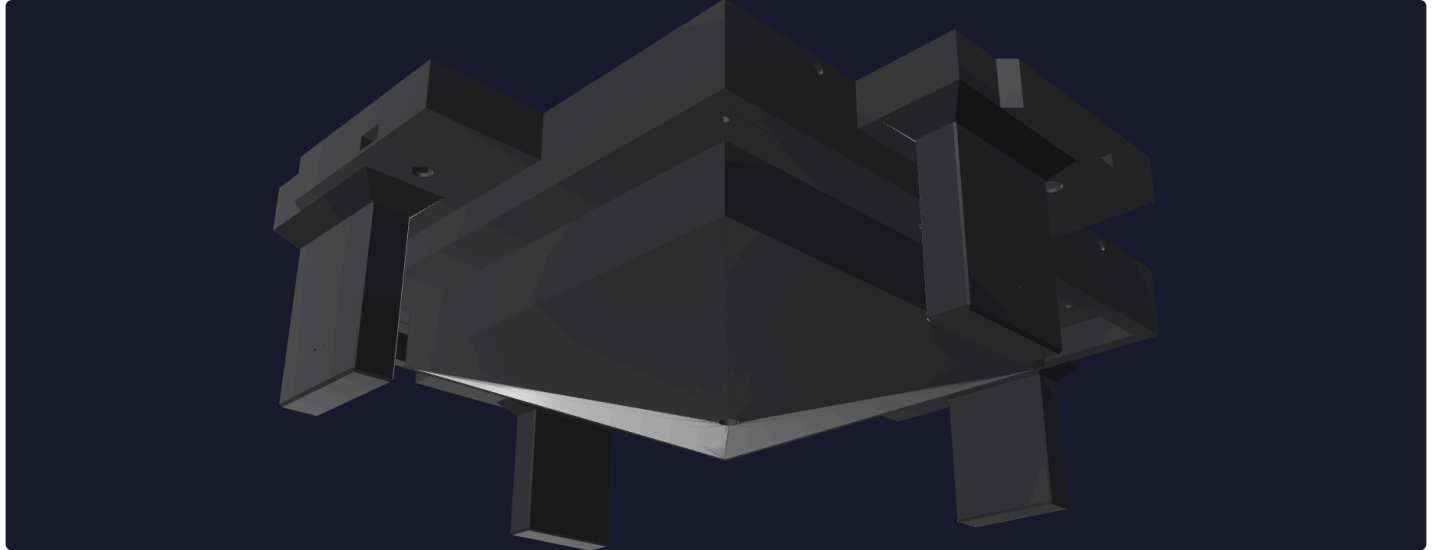
**1,271 g** on a 1 kg spool. Automatic backup, or a supervised change at around hour 30.

### DONE WHEN

The cavity is off the bed, cooled on the plate first, with both registration lands flat and no rock.

# The core

The half that drops in. It prints **upside down** — open back on the bed, forming plug and guide blades pointing up. **31 h 08 m · 1,016 g · 424 layers**.



The core seen from underneath: the **forming plug** that makes the inside of the funnel, the perimeter **plate** that forms the brim and carries the ports, four **lifting arms** and the four **guide blades** hanging below the parting plane.

Printing it inverted is what puts solid material where the load goes. The **8.4 mm** roof each square nut lifts against is laid down first, straight onto the bed; the thin lip that retains the nut is the bridged feature, not the load path. The blind cap over the rod socket bridges the same way, into the open back where you can see it.

## WHERE THE BRIDGING IS

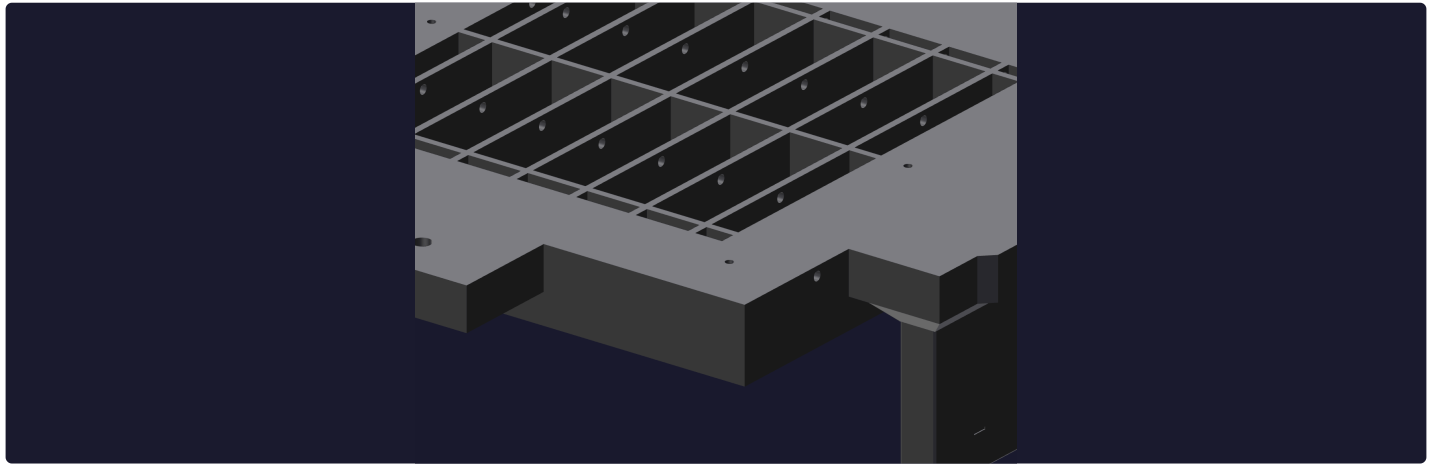
This plate carries almost all the bridging in the job: the nut-slot retaining lips and the blind rod-socket cap. Both bridge *away* from the forming surface, so a sagged underside is visible from the open back and does not touch the silicone.

## DONE WHEN

The core is off the bed with the skirt seating flat on a surface plate, no rock, and all four blades intact at their roots.

# Clear every passage

The mold breathes through small holes. A string left in one of them turns a vented mold into a sealed one, and you find out under vacuum with silicone already in it. Do this with a bright light and a piece of wire, on both halves, before anything else.



Rib bays in the core's open back, with the transverse **3 mm** backing-air holes through the rib walls. The same arrangement runs across the cavity near its feet.

- ☐ **Four transverse backing-air channels per half.** Cavity: near the feet. Core: near the open back. They must pass right through, side to side.
- ☐ **Four side air exits** at the feet of the cavity's guide towers.
- ☐ **Every rib bay** on both halves — strings, blobs, stray purge.
- ☐ **The pour throat** in the core plate, **11 mm**, and its dish.
- ☐ **Five silicone vents**, **2.5 mm** each. These print near the low end of what a 0.8 mm nozzle resolves; expect to open them.

## ⊖ TWO AIR SYSTEMS THAT NEVER MEET

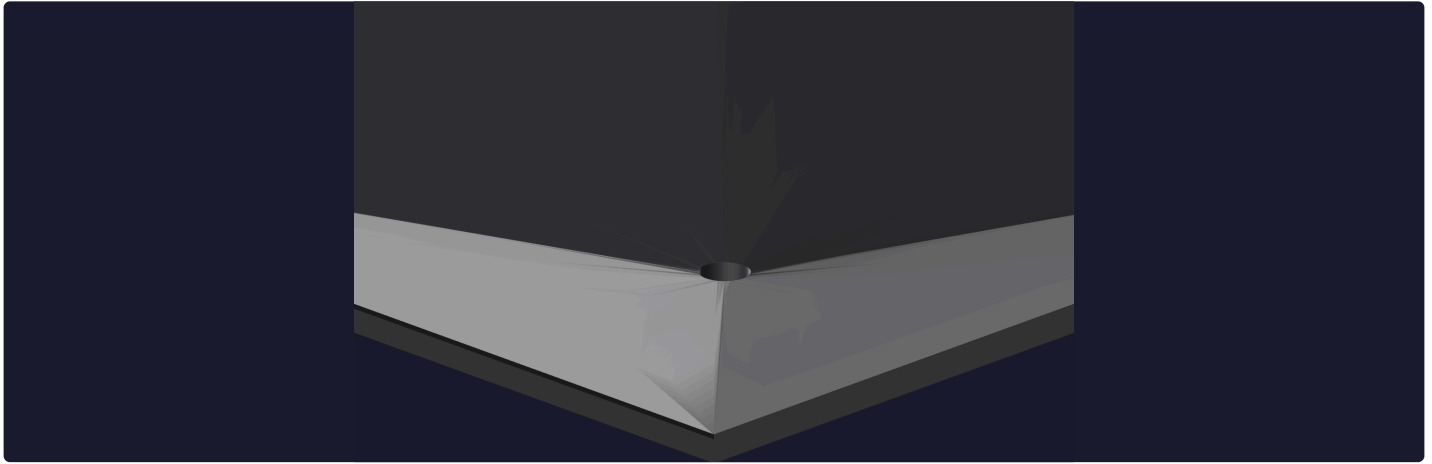
**Backing air** vents the structure to the chamber and must stay free of coating and silicone forever. **Pour and vents** carry silicone and must stay open through the cast. They do not connect — the geometry asserts it — and nothing you do should join them.

## DONE WHEN

Light passes through every backing channel and every tower exit, and a wire passes the pour throat and all five vents.

# Open the rod socket

The rod socket in the core is modelled at **6.45 mm** for a **6.35 mm** rod — **0.1 mm** of clearance, which is nothing. As printed, the *entry lip* comes out tighter than the rod. Expect to clear it. This is a step, not a defect.



The core's forming plug from below. Every surface of the funnel's floor falls to this one hole, and the hole is the mouth of the rod socket.

- 1 **Try the actual rod first.** Not a drill bit, not a gauge pin — the dowel you will cast with.
- 2 **Clear the entry lip.** Take the constriction at the mouth back to the socket's own diameter. Work from the mouth inward only.
- 3 **Use a depth stop.** The socket bottoms at **28.8 mm** and is **blind**. The closed end is a modelled backing cap; break through it and the core is scrap.
- 4 **Confirm the seating.** The rod must slide in by hand and bottom at 28.8 mm, leaving **22.0 mm** standing proud. Withdraw and re-seat it a few times; it should not bind at any depth.

## 🛑 NOMINAL, AND IT STAYS NOMINAL

This socket takes no coating, no primer and no release, ever. The rod is the funnel's entire spout bore, ground to a ten-thousandth — anything you add here you cast into the part.

## DONE WHEN

The rod slides freely to a hard stop at 28.8 mm, the blind end is intact, and 22.0 mm stands out below the plug.

# Dry-assemble, before any coating

The coupon proved four fits one at a time. It could not prove that four guides **238 mm** apart line up with four bearings. Only the real parts can, and only while you can still fix them.

- 1 **Put the rod in the core.** Seated, bottomed, 22 mm proud.
- 2 **Lower the core onto the cavity by hand.** The chamfered blade tips enter their bearings *before* the forming plug enters the cavity — that ordering is the whole reason the blades are long. Let it find its own way down; do not steer it.
- 3 **Check the skirt seats flat.** The core's skirt registers over the cavity for **10 mm**. It should sit down all the way round with no gap and no rock.
- 4 **Run the full stroke by hand.** Lift the core **32 mm** and set it back down, twice. Nothing should bind at any point in the travel.

## ⚠ NEVER JACK A TIGHT FIT

The screw jacks are lifting devices for a cured casting. If a guide binds or the rod is stiff, the jacks will happily bend a printed ear before the obstruction gives. Fix the fit with abrasive and inspection, at zero load, now.

## WRITE DOWN WHAT YOU LEARN

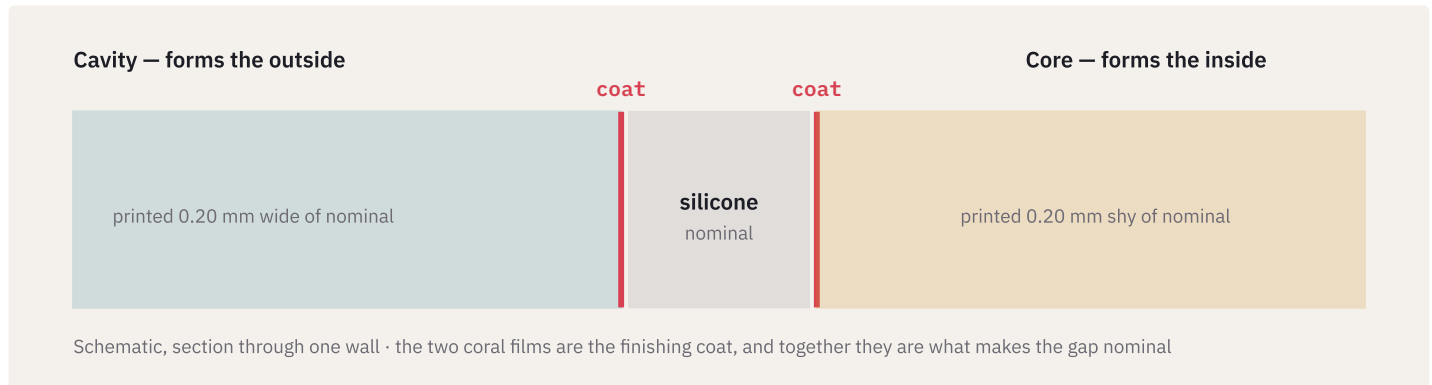
If one corner sits proud, or one blade enters later than the others, note which. After coating, the same symptom will mean something different — and you will want to know whether it was there beforehand.

## DONE WHEN

The core drops onto the cavity under its own weight, the skirt seats flat all round, and the full 32 mm stroke runs free with the rod installed.

# What 0.20 mm buys

Both forming faces are printed *away* from the funnel's real size. The cavity is grown **0.20 mm** everywhere; the core's plug is shrunk by the same. Finishing puts that thickness back as coating. The mold is only the right size once it is coated.



Net growth is **dry coating left, minus PETG removed by sanding**. It is a target you measure on the coupon, not a property of a can of anything.

## THE ARITHMETIC THAT CATCHES PEOPLE

A **0.20 mm** film on both walls of a channel closes it by **0.40 mm** ; on both sides of a boss it opens it by **0.40 mm** . Record the cavity's collar width and the core's width at matching stations **before** you coat, and you can read the film directly afterwards.

# Prove the process on the coupon

You are not choosing a coating out of a catalogue. You are proving that **this PETG, this coat, this release and this silicone** work together, on a 60 mm coupon, before you commit two parts that took 69 hours to print.

- 1 **Abrade, clean, dry.** Lightly — you are keying the surface, not removing the terraces yet.
- 2 **Coat the flat, the ramp and a vertical face.** Use a sandable coat and let it cure *fully*. A soft film sands into a wave.
- 3 **Sand and polish back to the datum.** Measure from the same stations as before. Target **0.20 mm ± 0.05** on the flat; record the vertical face separately; look along the ramp for pooling at its foot.
- 4 **Then cast against it.** Release the coupon, put silicone on it, cure it, pull it off. Do that **more than once on the same coupon**.

## What the cast half of the trial has to show

- The silicone **cures through**, right down to the coated face — no tack, no soft skin.
- It **releases cleanly**, without taking coating with it.
- The coating **survives repeated demolding**, not just the first one.
- Bubbles stop. Put a small coated patch under the silicone and watch for gas still coming out of the print. Ventilation does not make a printed mold airtight; the continuous coating is what does.

### THE GUIDE'S CALL, NOT SPEC

The shop holds two candidate coats — Krylon Crystal Clear acrylic and Craft Resin epoxy. The acrylic is the one already chosen elsewhere in this machine for a silicone contact face, because it seals print porosity and does not inhibit a platinum cure. **Start there.** The trial is what decides it, and the trial is cheap.

### ⚠ SOLVENT AND AEROSOL

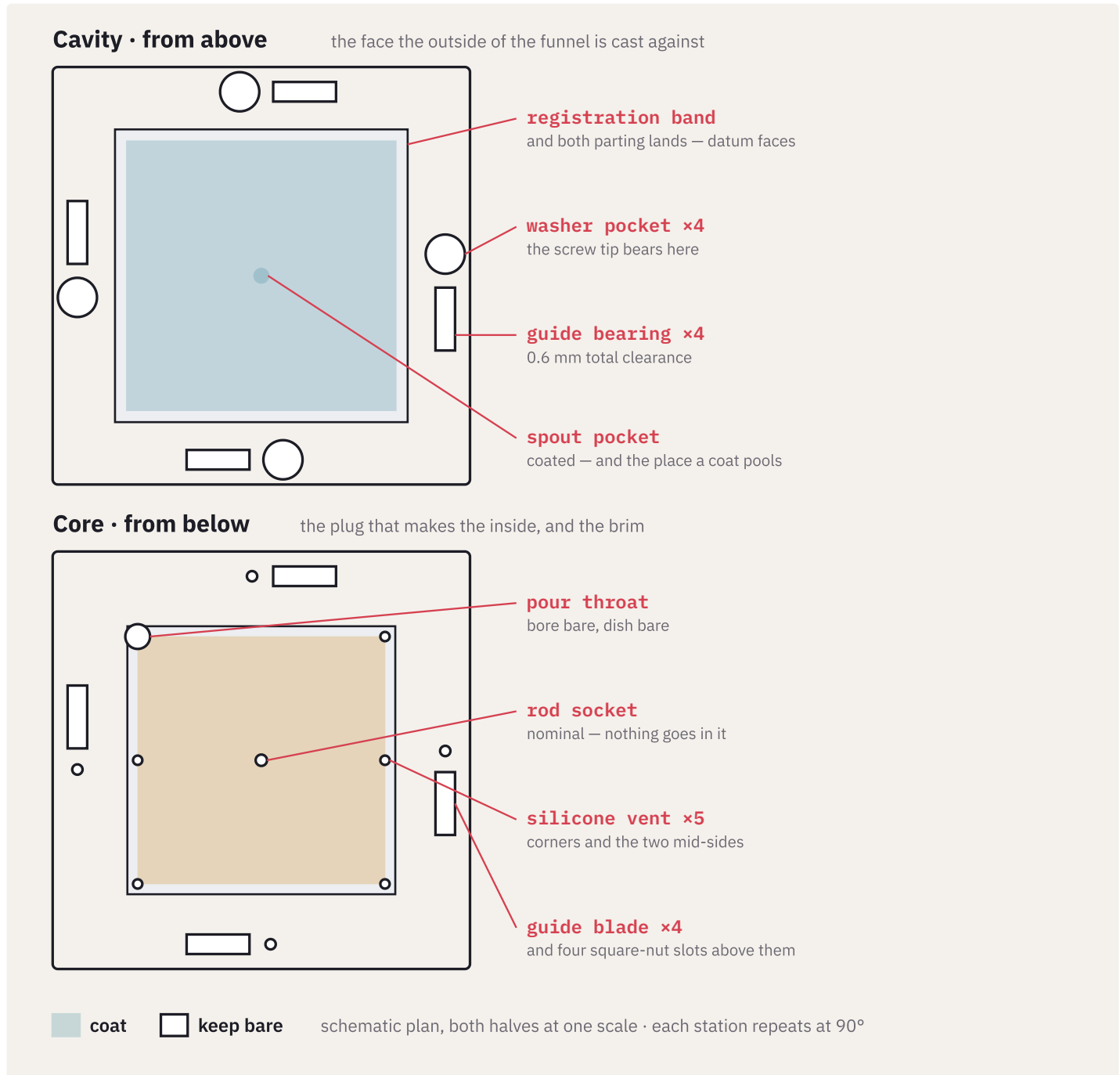
Spray coats and their thinners: ventilate, and keep them away from the silicone stock.

### DONE WHEN

One coupon reads  $0.20 \pm 0.05$  mm on the flat and has released cured silicone cleanly at least twice.

# Two faces get coated

Everything the silicone touches gets the film. Everything that slides, seats, registers or breathes stays bare. Mask against this page, not against memory.



Coat the cavity's bowl and the core's plug, out to the edge of the forming face — and stop there. The brim recess in the core plate's underside is forming surface too.



# Mask, then coat

Work through this list on both halves and tick it off. Anything on it that takes a film is a fit you cannot get back without stripping the part.

## MASK BEFORE YOU SPRAY

- ☐ **The rod socket.** 0.1 mm of clearance. Plug it.
- ☐ **Four guide blades and four bearings.** 0.6 mm total across a 12 mm width.
- ☐ **Four square-nut slots** and their bores.
- ☐ **Four washer pockets.** A film under a washer is a soft bearing.
- ☐ **The registration skirt and both parting lands.**
- ☐ **Every backing bay, air channel and tower exit.**
- ☐ **The pour throat and all five vent bores.**

## THEN COAT

- 1 **Apply the proven process to the two forming faces.** The cavity's bowl and the core's plug, including the brim recess in the plate's underside.
- 2 **Work up to the forming-face edges, and stop.** No bead on the parting land — a bead there holds the mold open and floods the brim.
- 3 **Do not round the trim shoulder.** The 1 mm step at the spout exit is what your blade indexes on after the cast. Round it now and you will be cutting the funnel's bore freehand.
- 4 **Watch the spout pocket.** Deep, narrow, and the one place a coating pools instead of levelling. Inspect it under light before you call the coat done.
- 5 **Cure fully and dry.** Then sand and polish to the number the coupon gave you.

### 🚫 THE COAT IS THE SEAL

Printed road-to-road porosity is what makes a printed mold leak gas into silicone. Both forming faces need a **continuous** film. A thin patch is not a cosmetic problem; it is where the bubbles come from.

### DONE WHEN

Both forming faces carry a continuous, fully cured, polished film, and every item in the mask list is still bare PETG.

# Read the film on the real parts

The coupon measured your *process*. It cannot tell you the film is even over a 270 mm part. The before-and-after widths can, and they cost you nothing but writing numbers down.

| STATION                | BEFORE COATING | AFTER COATING | EXPECTED CHANGE           |
|------------------------|----------------|---------------|---------------------------|
| Cavity collar width, X | —              | —             | internal, <b>-0.40 mm</b> |
| Cavity collar width, Y | —              | —             | internal, <b>-0.40 mm</b> |
| Core plug width, X     | —              | —             | external, <b>+0.40 mm</b> |
| Core plug width, Y     | —              | —             | external, <b>+0.40 mm</b> |
| Brim recess depth      | —              | —             | <b>-0.20 mm</b>           |

Measure at matching stations on both halves, so a difference is a film and not two different places.

## Then compare against the funnel

Put the finished faces against the nominal funnel geometry: the brim thickness, the ramp transitions where the floor meets the walls, and the collar. What you are looking for is not a single number in tolerance — it is the film being *the same* in the bowl, on the walls and in the spout pocket.

### THE LAST WORD BELONGS TO THE FIRST CAST

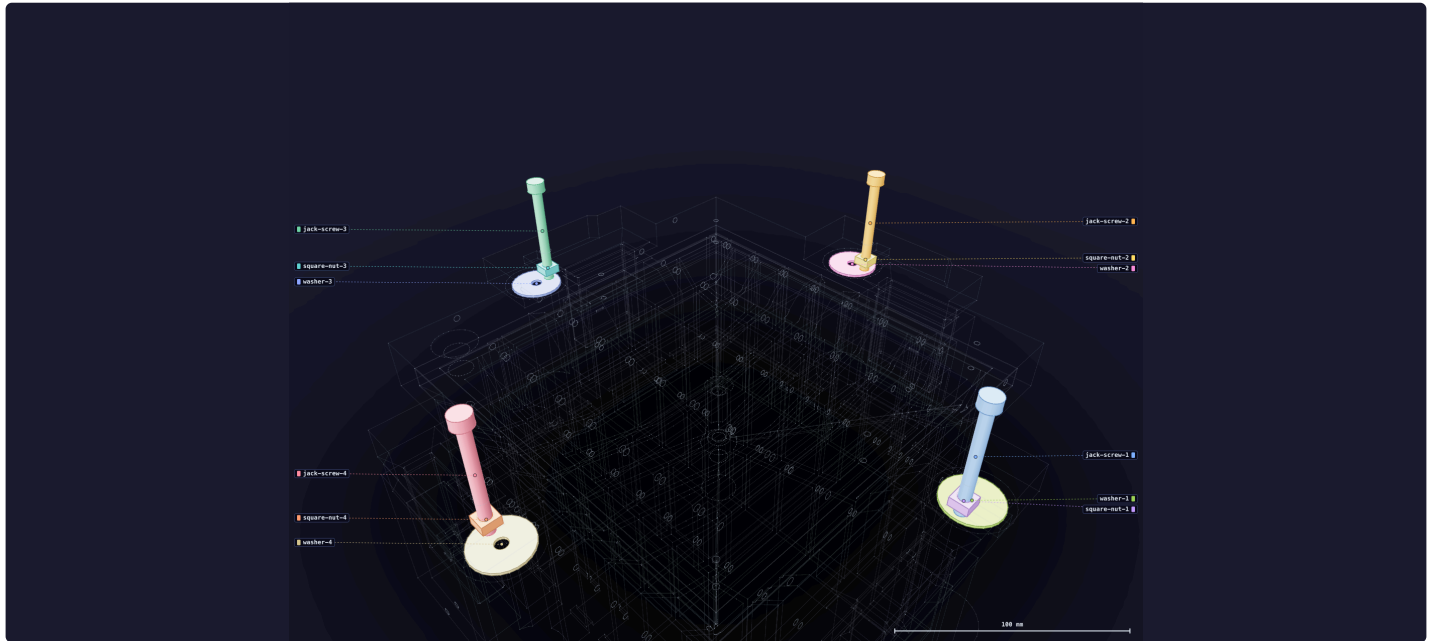
No amount of measuring the tooling settles this. The first funnel out of the mold is the real dimensional and surface check — treat it as a test piece and expect to learn something from it.

### DONE WHEN

Before and after widths are recorded at matching stations on both halves, and the change at each is within  $\pm 0.05$  mm of the expected 0.40.

# Fit the four jacks

One station repeats at each side of the mold. A screw threads through a square nut captured in the core's arm and rests its tip on a steel washer sitting in the cavity. Turning it down cannot push the washer anywhere, so the core goes up instead.



All four stations, with the mold ghosted. Screw, nut and washer at each; the blade that keeps the core square runs alongside, inboard of the screw.

- A** Slide one square nut into each core arm from its outboard edge, into the side-entry slot.
- B** Pass a screw down through the arm and into the nut to retain it. The nut is captured by the screw, not pressed in — it is meant to be a little loose until then.
- C** Lay one washer flat in each cavity pocket. The screw tip lands on the solid ring, not the hole; the axis is deliberately off-centre.
- D** Back all four screws out until their tips clear the washers with the core seated.

## KEEP THE SCREW ENDS FLAT

A burr on a screw tip ploughs the washer, and a deeply scored washer gets replaced rather than reused. Check the four tips before they ever touch steel.

## DONE WHEN

Four nuts captured, four washers flat in their pockets, four screws backed clear, and the core still seats fully with all four fitted.

# Dry-run it in the chamber

The last thing before silicone. Assemble the mold empty, put it in the vacuum chamber, pull it down and let it back up. It costs ten minutes and it is the only way to find a coating that lifts or a channel that closed, while the mold is still clean.

- 1 **Measure the chamber, not the spec.** The mold's widest points are the cavity's guide-tower corners. The nominal clearance to the chamber wall is about **7 mm** on the radius — and your catch tray, its rim, and whatever holds the core down all have to live inside that too.
- 2 **Stand it so the backing-air exits stay open.** A flat shelf under the cavity, or a clamp board over the core, still leaves them breathing; a blanket or a soft mat does not.
- 3 **Pull down, hold, admit air slowly.**
- 4 **Then take it apart and look.** Distortion at the skirt or the parting land. Coating lifted, crazed or blistered. Any channel or port that has closed. Run the 32 mm stroke again and confirm nothing has changed.

## THE GUIDE'S CALL, NOT SPEC

Nothing in the source specifies what holds the core down during the cast — only that something must, that the jacks are lifting devices and do not clamp, and that the backing-air exits must stay uncovered. A flat board across the core's four arms with dead weight on it satisfies all three, keeps the load on the arms rather than the plate, and leaves the ports and the open back clear. Work that out now, on the dry run, with the chamber lid closing over it.

## DONE WHEN

The assembled mold has been through a full vacuum cycle, comes apart unchanged, and its holding arrangement fits in the chamber with the lid closed.

# You have a tool

Two coated halves, four steel stations, one ground rod, and a coupon with numbers written on it. The mold is reusable; from here on the work is casting, and that is the other document.

## What this guide does not decide

These are open in the source, not answered here. Each one is a thing you will meet at the bench, and each one is worth writing down once you have settled it.

- **Which coating.** Two candidates in the shop, one purpose-made alternative not owned. The coupon trial decides it; nothing else can.
- **How many coats, and what grit.** "Lightly abrade" and "sand and polish" are the whole specification. Your coupon numbers are the answer.
- **The holding arrangement.** Named four times in the source, specified nowhere.
- **The catch tray.** Required under the mold, no object named.
- **How the mask is made** — tape, liquid mask, plugs in the bores.

### KEEP THE COUPON

Do not throw the finish witness away when the mold is built. It is the record of what your process produced, and the reference when you re-coat a worn face later.

Next: **Casting a funnel** — mix, degas, pour, close, vacuum, cure, jack, peel, trim, post-cure.